Supporting Information for

Stable Surfaces that Bind too Tightly: Can Range Separated Hybrids or DFT+U Improve Paradoxical Descriptions of Surface Chemistry?

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Table S1. Geometry-optimized lattice parameters *a* and *c* in Å of rutile-type transition metal oxides MO_2 (M=T, V, Mo, Ru, Rh, Ir, Pt) with percentage of HF exchange. The experimental lattice parameters (exp) are also shown.

M		exp	PBE	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
Ti	*a*	4.594	4.620	4.608	4.596	4.585	4.574	4.564	4.554	4.544	4.535	4.526	4.518
	c	2.959	2.983	2.981	2.979	2.977	2.976	2.974	2.973	2.972	2.971	2.971	2.970
V	*a*	4.555	4.570	4.559	4.551	4.549	4.567	4.557	4.548	4.501	4.497	4.492	4.519
	c	2.853	2.795	2.785	2.769	2.745	2.691	2.685	2.680	2.734	2.721	2.711	2.657
Mo	*a*	4.847	4.980	4.973	4.967	4.960	4.954	4.948	4.942	4.938	4.932	4.927	4.920
	c	2.814	2.750	2.733	2.717	2.704	2.690	2.679	2.669	2.657	2.650	2.642	2.636
Ru	*a*	4.492	4.608	4.584	4.562	4.543	4.526	4.511	4.499	4.487	4.475	4.466	4.458
	c	3.107	3.163	3.159	3.155	3.152	3.150	3.148	3.148	3.152	3.157	3.159	3.156
Rh	*a*	4.489	4.620	4.598	4.580	4.563	4.548	4.533	4.520	4.508	--	--	--
	c	3.090	3.182	3.165	3.155	3.144	3.135	3.125	3.118	3.112	--	--	--
Ir	*a*	4.505	4.522	4.508	4.495	4.482	4.472	4.459	4.448	4.437	4.432	4.421	--
	c	3.159	3.200	3.189	3.180	3.171	3.161	3.154	3.146	3.139	3.128	3.122	--
Pt	*a*	4.485	4.623	4.606	4.591	4.577	4.564	4.550	4.539	4.526	4.515	4.504	4.498
	c	3.130	3.322	3.288	3.255	3.227	3.208	3.194	3.183	3.173	3.164	3.155	3.144

Table S2. Geometry-optimized lattice parameters *a* and *c* in Å of rutile-type transition metal oxides MO_2 (M=T, V, Mo, Ru, Rh, Ir, Pt) with U in (eV) for DFT+U. The experimental lattice parameters (exp) are also shown.

M		exp	PBE	1 eV	2 eV	3 eV	4 eV	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
Ti	*a*	4.594	4.614	4.617	4.621	4.624	4.627	4.631	4.635	4.641	4.649	4.657	4.667
	c	2.959	2.955	2.968	2.983	2.999	3.016	3.034	3.053	3.073	3.093	3.113	3.133
V	*a*	4.555	4.595	4.597	4.600	4.612	4.635	4.648	4.659	4.671	4.686	4.703	4.724
	c	2.853	2.736	2.738	2.742	2.749	2.757	2.765	2.771	2.778	2.788	2.800	2.817
Mo	*a*	4.847	4.962	4.962	4.962	4.962	4.962	4.962	4.962	4.962	4.962	4.962	4.962
	c	2.814	2.742	2.733	2.723	2.714	2.706	2.698	2.691	2.685	2.678	2.674	2.666
Ru	*a*	4.492	4.642	4.634	4.627	4.622	4.619	4.616	4.614	4.613	4.613	4.612	4.612
	c	3.107	3.187	3.187	3.188	3.189	3.189	3.190	3.190	3.191	3.191	3.191	3.192
Rh	*a*	4.489	4.625	4.625	4.626	4.626	4.627	4.629	4.630	4.632	4.633	4.634	4.635
	c	3.090	3.171	3.169	3.168	3.166	3.165	3.164	3.163	3.162	3.161	3.160	3.160
Ir	*a*	4.505	4.577	4.577	4.578	4.578	4.578	4.579	4.579	4.579	4.580	4.581	4.582
	c	3.159	3.212	3.209	3.205	3.202	3.200	3.197	3.195	3.192	3.190	3.188	3.187
Pt	*a*	4.485	4.632	4.631	4.631	4.631	4.630	4.629	4.629	4.628	4.627	4.626	4.625
	c	3.130	3.280	3.276	3.272	3.268	3.266	3.264	3.263	3.263	3.262	3.262	3.261

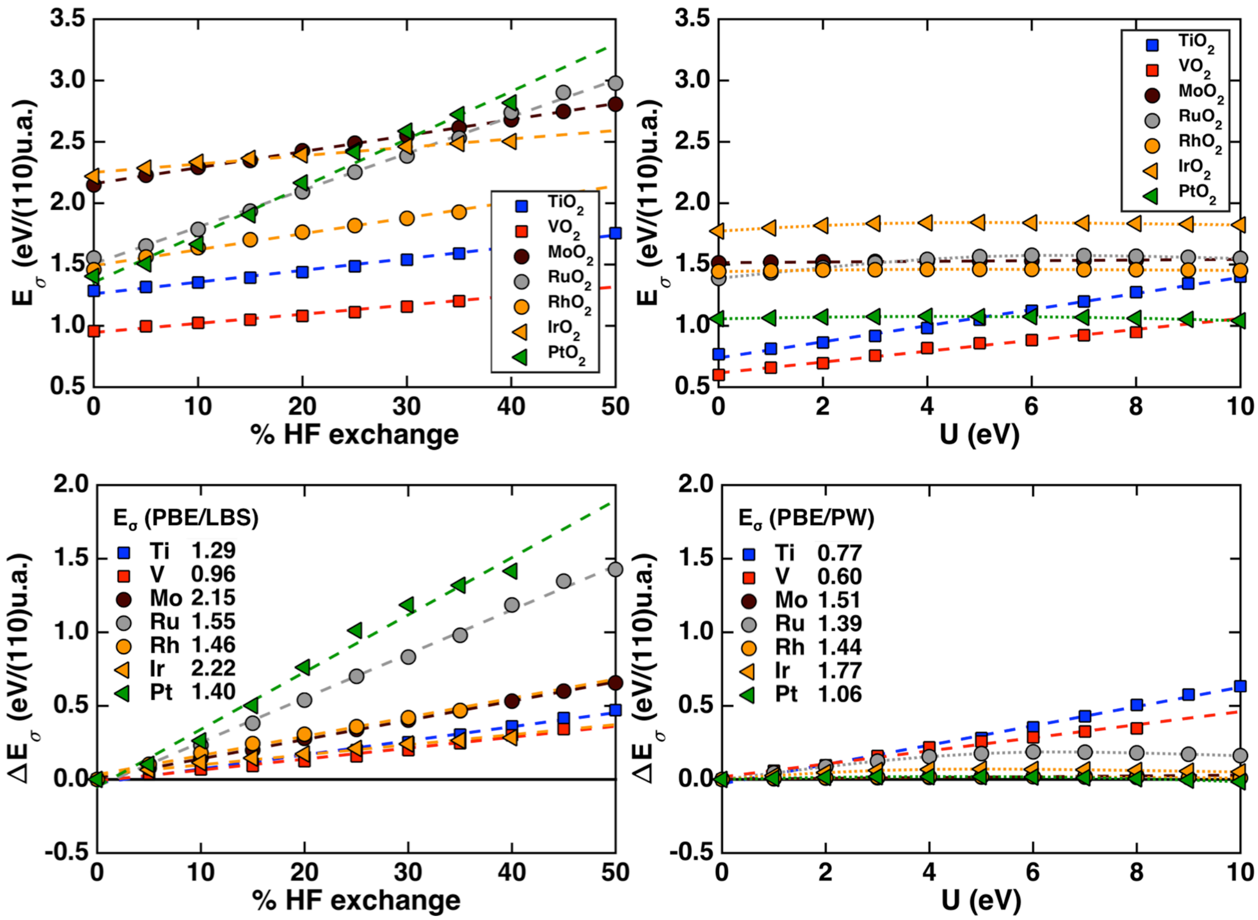

Figure S1. Dependence of (top) surface formation energies, E_σ, and (bottom) shift of surface formation energies, ΔE_σ, in eV/(110) unit area (u.a.) on (left) percentage of HF exchange and (right) *U* (in eV) for nonmagnetic rutile-type MO_2 (110) surface with M = Ti (blue squares), V (red squares), Mo (brown circles), Ru (gray circles), Rh (orange circles), Ir (orange triangles), and Pt (green triangles). For surface formation energies, the GGA surface formation energy is indicated in the inset, with all curves aligned to have zero ΔE_σ values at the pure GGA value. The dashed lines indicate linear best fits through all data to quantify the surface formation energy sensitivities to HF exchange or *U*. Both HF exchange and Hubbard *U* y-axes span the same 3.0 eV/(110)u.a. range in top figures and 2.5 eV/(110)u.a. range in bottom figures.

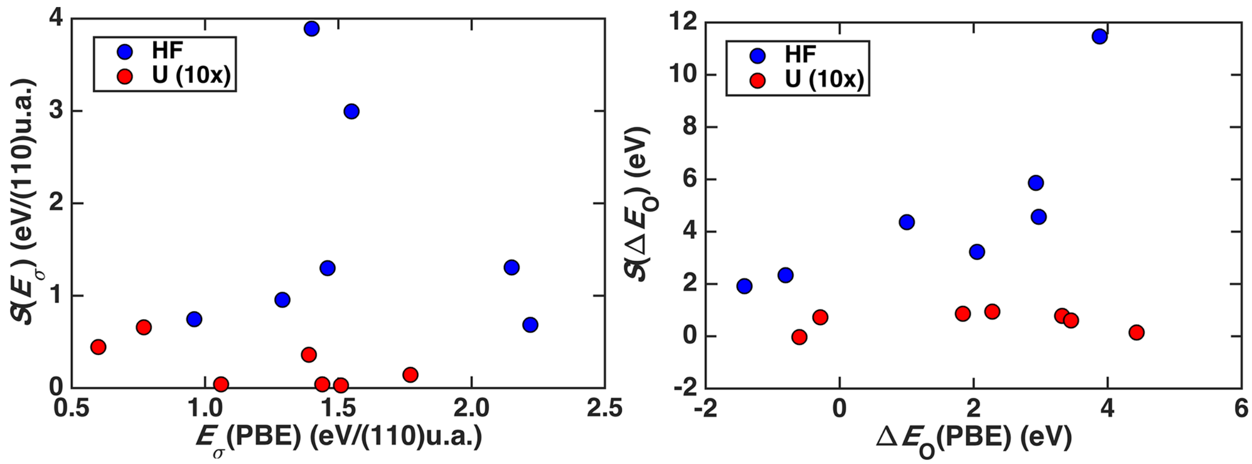

Figure S2. (left) Surface formation energy sensitivities versus surface formation energy evaluated at pure PBE-GGA limit for HF exchange tuning in an LBS (blue circles) or for DFT+U tuning with in a PWBS (red circles). (right) Oxygen adsorption energy sensitivities versus oxygen adsorption energy evaluated at pure PBE-GGA limit for HF exchange tuning in an LBS (blue circles) or for DFT+U tuning in a PWBS (red circles).

Table S3. Bond lengths between metal adsorption site and O-atom adsorbate in Å with increasing HF exchange. Missing values correspond to calculations at high HF exchange that did not converge.

	PBE	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
TiO_2	1.716	1.718	1.720	1.722	1.728	1.808	1.919	1.999	2.082	2.139	2.198
VO_2	1.586	1.588	1.589	1.590	1.592	1.594	1.595	1.597	1.598	--	--
MoO_2	1.725	1.727	1.730	1.732	1.734	1.737	1.740	1.741	1.743	--	--
RuO_2	1.761	1.765	1.770	1.775	1.781	1.784	1.789	1.792	1.794	--	--
RhO_2	1.818	1.820	1.823	1.825	1.828	1.831	1.833	--	--	--	--
IrO_2	1.789	1.793	1.798	1.804	1.807	1.811	1.816	1.820	--	--	--
PtO_2	1.855	1.856	1.858	1.860	1.861	1.864	1.868	1.869	--	--	--

Table S4. Bond lengths between metal adsorption site and O-atom adsorbate in Å with increasing *U* values. Missing values correspond to calculations at high *U* value that did not converge.

	PBE	1 eV	2 eV	3 eV	4 eV	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
TiO_2	1.676	1.679	1.682	1.687	1.693	1.701	1.719	1.752	1.790	1.825	1.852
VO_2	1.589	1.589	1.590	1.591	1.592	1.593	1.593	1.594	1.595	--	--
MoO_2	1.704	1.705	1.705	1.706	1.706	1.707	1.708	1.709	1.709	--	--
RuO_2	1.809	1.809	1.810	1.811	1.811	1.812	1.813	1.814	1.815	1.815	1.816
RhO_2	1.825	1.826	1.827	1.829	1.830	1.832	1.834	1.836	1.838	1.841	1.843
IrO_2	1.814	1.814	1.814	1.814	1.814	1.814	1.814	1.815	1.815	1.816	1.816
PtO_2	1.862	1.864	1.867	1.870	1.873	1.876	1.879	1.881	1.883	1.885	1.887

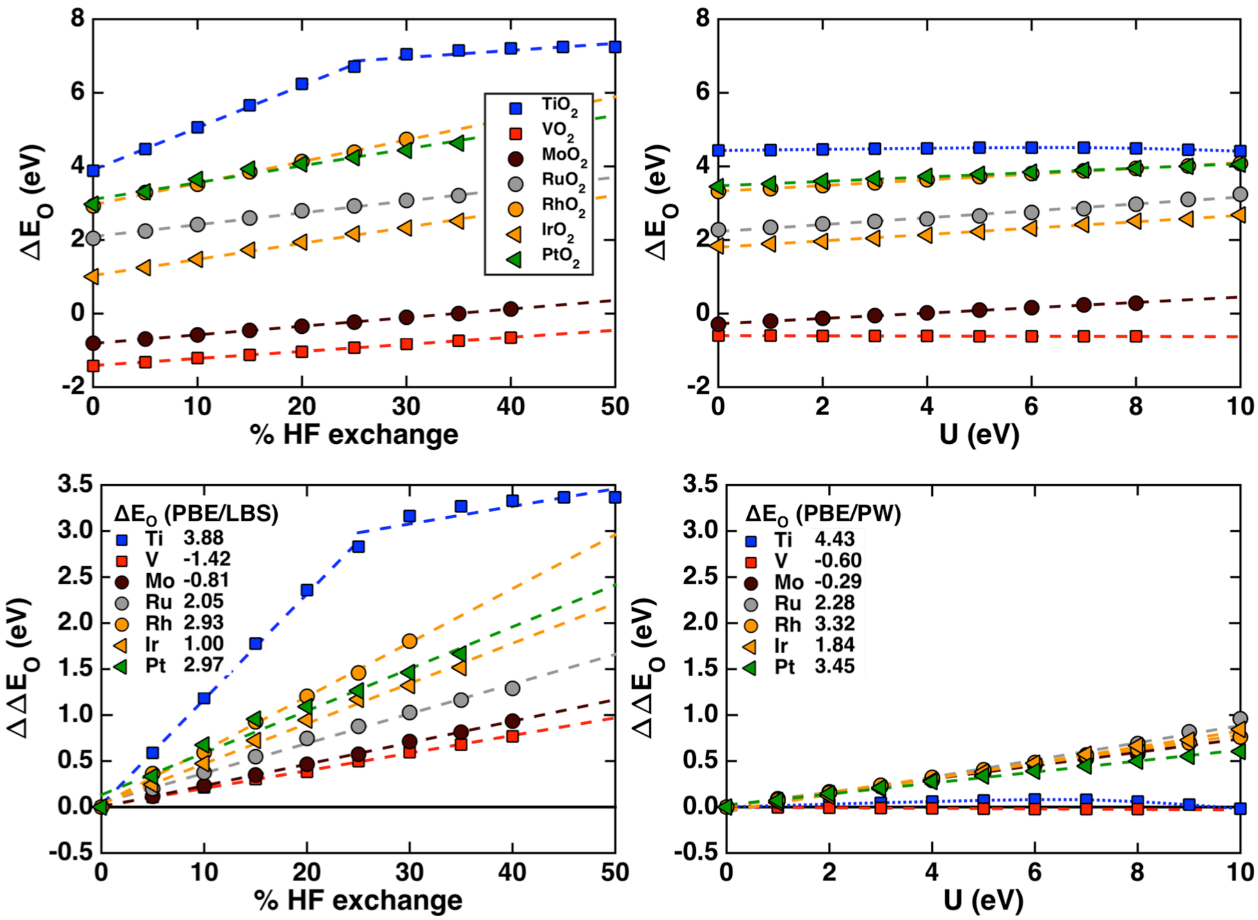

Figure S3. Dependence of (top) oxygen adsorption energies, ΔE_O, and (bottom) shift of oxygen adsorption energies, $\Delta\Delta E_O$, in eV on (left) percentage of HF exchange and (right) *U* (in eV) for nonmagnetic rutile-type MO_2 (110) surface with M = Ti (blue squares), V (red squares), Mo (brown circles), Ru (gray circles), Rh (orange circles), Ir (orange triangles), and Pt (green triangles). For shift of adsorption energies, the GGA oxygen adsorption energy is indicated in the inset, with all curves aligned to have zero $\Delta\Delta E_O$ values at the pure GGA value. The dashed lines indicate linear best fits through all data to quantify the adsorption energy sensitivities to HF exchange or *U*. Both HF exchange and Hubbard *U* y-axes span the same 10 eV range in top figures and 4.0 eV range in bottom figures.

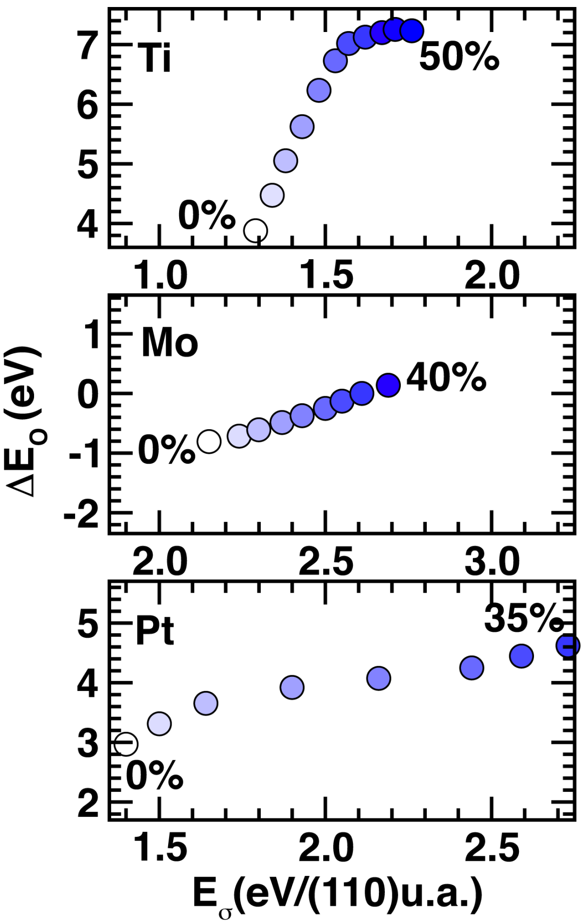

Figure S4. Relationship between ΔE_O (in eV) and E_σ (in eV/(110)u.a.) for an early (Ti), mid-row (Mo) and late (Pt) rutile transition metal oxide with varying HF exchange. The 0% point is colored in white up to 50% colored in blue. The Mo and Pt cases stop at 40% and 35%, respectively, because one of the two relevant calculations did not converge. All x- and y-axes span the same range but are shifted to center the relevant data.

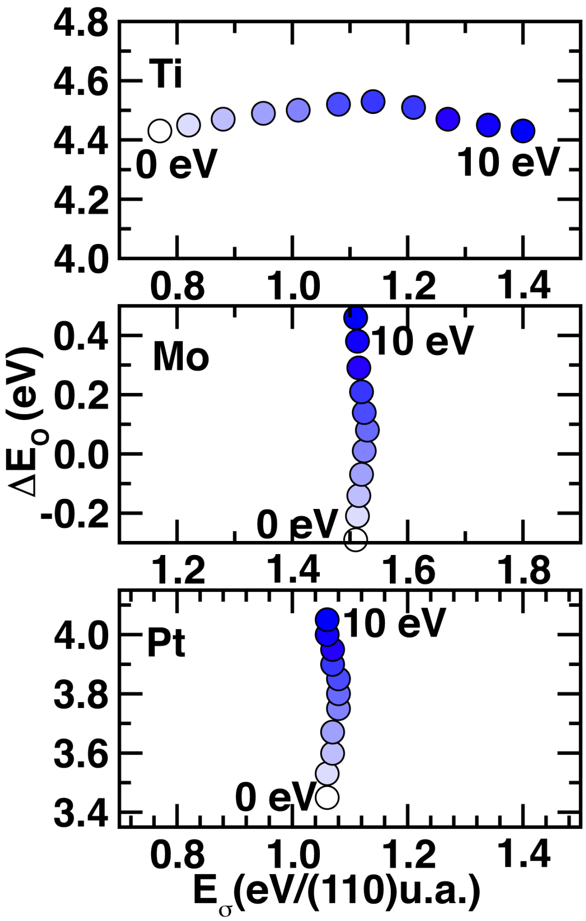

Figure S5. Relationship between ΔE_O (in eV) and E_σ (in eV/(110)u.a.) for an early (Ti), mid-row (Mo) and late (Pt) rutile transition metal oxide with +U correction. The $U = 0$ eV point is colored in white up to $U = 10$ eV colored in blue. All x- and y-axes span the same range but are shifted to center the relevant data.

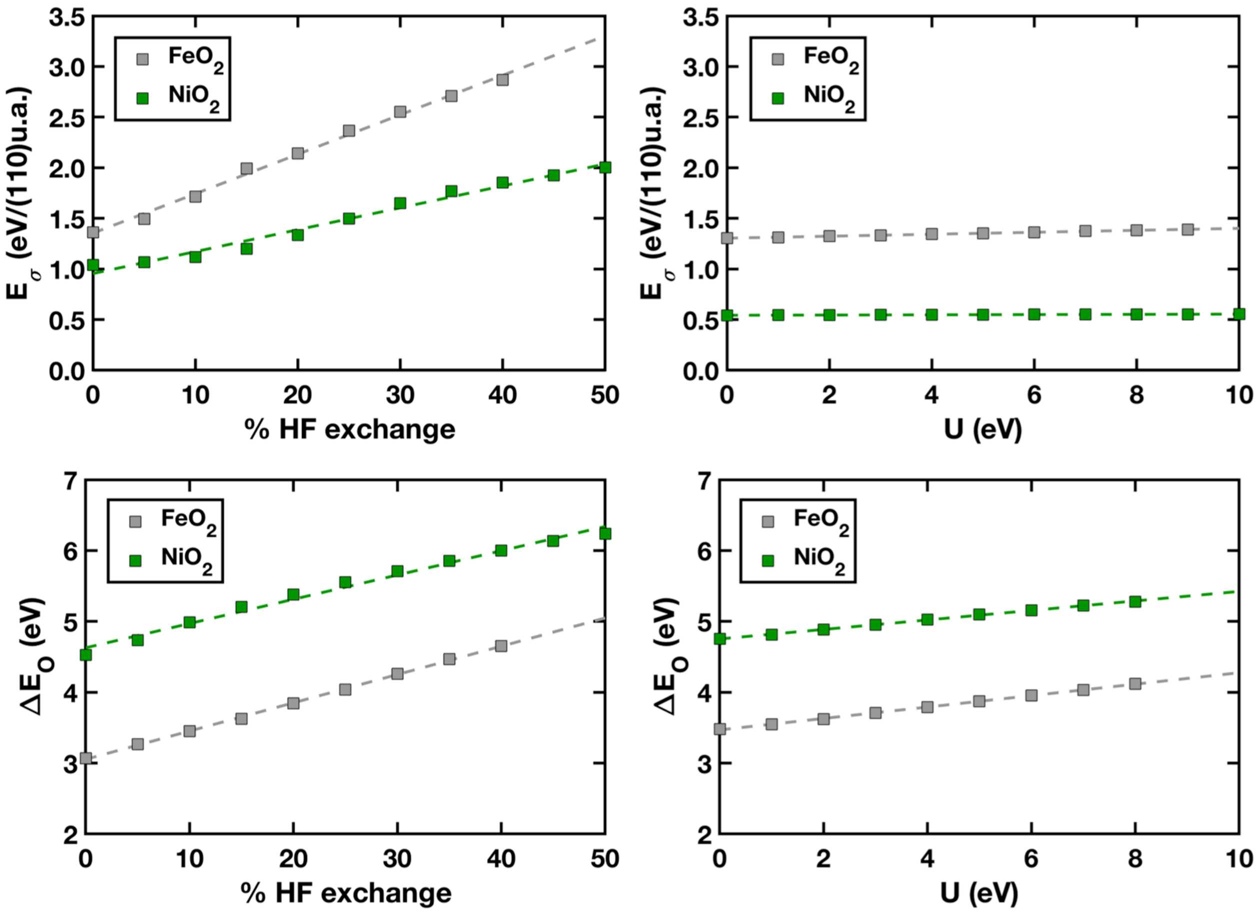

Figure S6. Dependence of (top) surface formation energies, E_σ, and (bottom) oxygen adsorption energies, ΔE_O, on (left) percentage of HF exchange and (right) *U* (in eV) for nonmagnetic rutile-type MO_2 (110) surface with M = Fe (gray squares), Ni (green squares). The dashed lines indicate linear best fits through all data to quantify the surface formation energies or adsorption energy sensitivities to HF exchange or *U*. Both HF exchange and Hubbard *U* y-axes span the same 3.5 eV/(110)u.a. range in top figures and 5 eV range in bottom figures.

Table S5. Sensitivities of surface formation energies and oxygen adsorption energies to percentage of HF exchange and *U* for FeO_2 and NiO_2.

	FeO_2 E_σ	NiO_2 E_σ	FeO_2 ΔE_O	NiO_2 ΔE_O
HF exchange	3.90	2.17	3.99	3.42
U (10x)	0.10	0.01	0.81	0.67

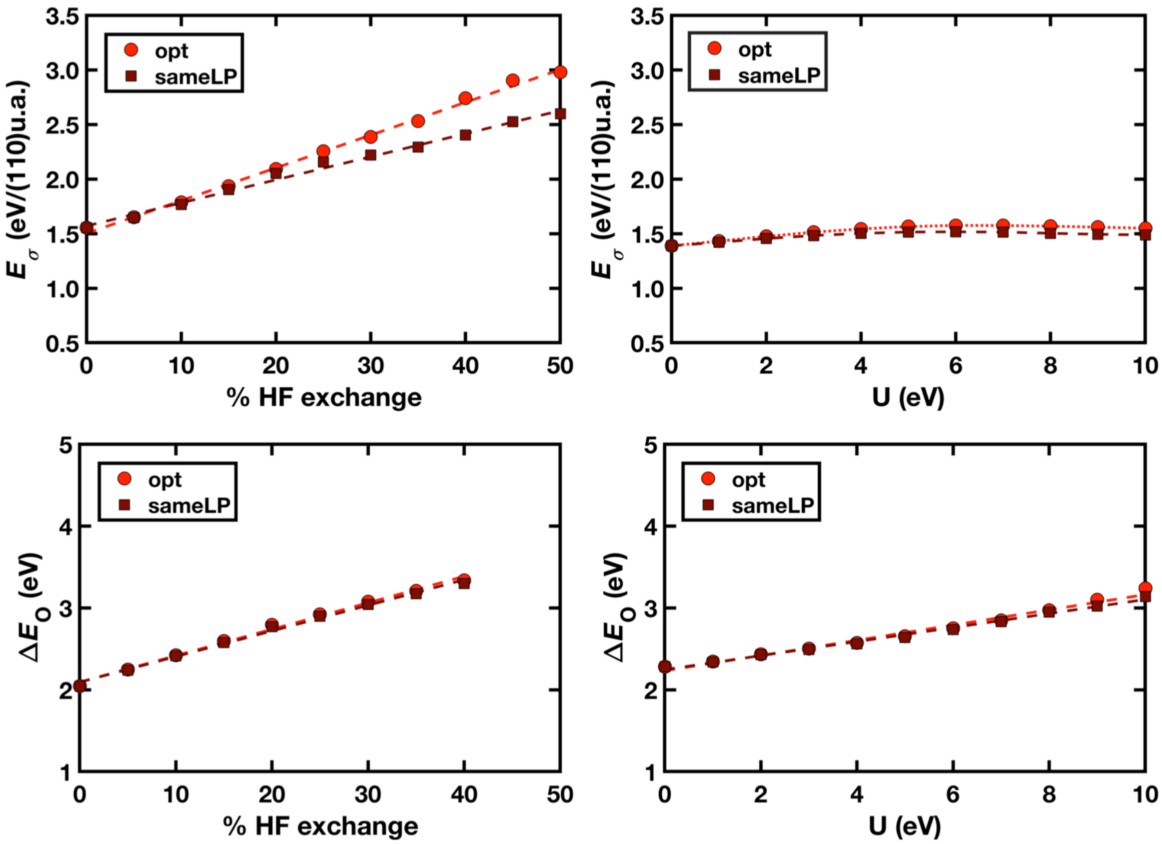

Figure S7. Comparison of surface formation energies, E_σ, in eV/(110) unit area (u.a.) (top), or oxygen adsorption energies, ΔE_O, in eV (bottom) with percentage of HF exchange (left) and U (in eV, right) for RuO_2 between geometry optimized lattice parameter (red circles) at each tuning parameter and fixed PBE-optimized lattice parameter (maroon squares). The dashed lines indicate linear best fits through all data to quantify the surface formation energy sensitivities or oxygen adsorption energy sensitivities to HF exchange or *U*. RuO_2 was selected since it exhibits the largest lattice constant deviation with HF exchange tuning, i.e., 0.15 Å.

Method S1. Details of DLPNO-CCSD(T), reference energy calculations.

DLPNO-CCSD(T)/aug-cc-pVTZ calculations were performed in Orca v4.0 on an octahedral transition metal complex, $[Ti(H_2O)_6]^{2+}$.

1. The surface formation energy, E_σ(DLPNO), was computed by i) geometry optimizing singlet $[Ti(H_2O)_6]^{2+}$ with PBE0/aug-cc-pVTZ, ii) computing DLPNO-CCSD(T)/aug-cc-pVTZ (Tight PNO threshold) single point, iii) moving an axial H_2O ligand to 5 Å away from the Ti center, iv) computing DLPNO-CCSD(T)/aug-cc-pVTZ (Tight PNO threshold) single point on the stretched geometry, and v) obtaining E_σ(DLPNO) from the difference of the two DLPNO-CCSD(T) single point energies. We also obtained the values for the same cluster with PBE0 (i.e., E_σ(PBE0)).

2. The oxygen adsorption energy, ΔE_O(DLPNO), was computed by i) extracting the TiO_5 structure (i.e., Ti and the O atoms coordinated from the material) from the PBE0 periodic oxygen-adsorption TiO_2 surface structure by keeping the surface Ti adsorption site and the five coordinated oxygen atoms, ii) H-capping to form a $[Ti(H_2O)_5]^{2+}$ complex, iii) geometry optimizing only the H-atoms with PBE0/aug-cc-pVTZ, iv) carrying out DLPNO-CCSD(T)/aug-cc-pVTZ single points with an O atom varied with Ti-O distance from 1.50 Å to 1.85 Å in increments of 0.01 Å to select the minimum energy structure, v) performing a single point

energy calculation with DLPNO-CCSD(T) at larger distance 5.00 Å, and vi) obtaining the oxygen adsorption energy as the energy difference between the last two quantities. We also obtained the values for the same cluster with PBE0 (i.e., ΔE_O(PBE0)).

3. We adjusted the DLPNO-CCSD(T) energies to be a correction to the periodic systems by obtaining the difference between the cluster and periodic systems with PBE0 as well as the difference between DLPNO-CCSD(T) and PBE, i.e., E_σ(DLPNO)+ E_σ(periodic)- E_σ(PBE0) and ΔE_O (DLPNO)+ ΔE_O(periodic) - ΔE_O (PBE0).

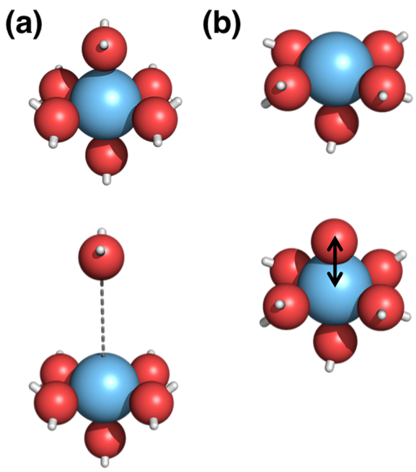

Figure S8. Models of octahedral transition metal complex $[Ti(H_2O)_6]^{2+}$ for DLPNO-CCSD(T) reference of (a) surface formation energies, and (b) oxygen adsorption energies.

Table S6. Surface formation energies and oxygen adsorption energies in eV from the DLPNO-CCSD(T) method and PBE0 functional on periodic system and molecular transition metal complex. The transition metal complex values are explicitly computed for both methods, and the periodic DLPNO-CCSD(T) value is obtained as a correction based on the difference of PBE0 and DLPNO-CCSD(T) in the complex.

	Transition metal complex		Periodic	
	PBE0	DLPNO-CCSD(T)	PBE0	*DLPNO-CCSD(T) (approx.)*
E_σ	1.474	1.427	1.485	*1.439*
ΔE_O	6.012	5.018	6.712	*5.718*

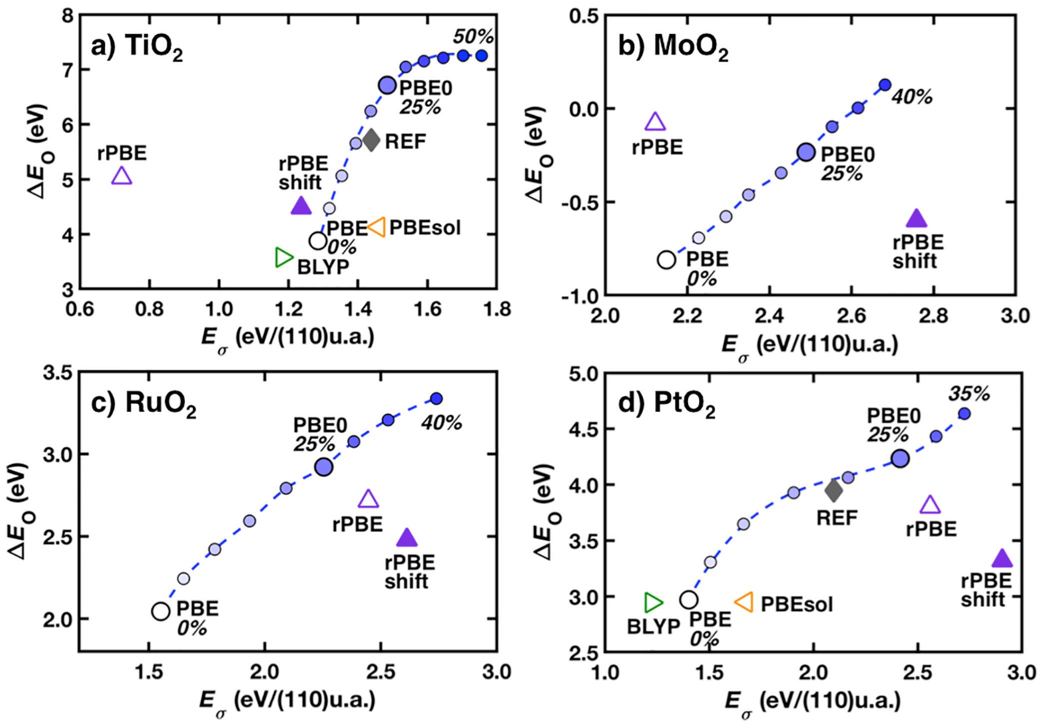

Figure S9. Surface formation energies, E_σ, in eV/(110) unit area (u.a.) versus oxygen adsorption energies, ΔE_O, in eV for α_{HF} tuning the PBE0 global hybrid functional, as well as the revPBE functional with plane wave basis set (rPBE, empty purple triangle), and the shift revPBE result (rPBE shift, purple triangle) obtained by adding the PBE result difference between localized basis set and plane wave basis set for a) TiO_2(110), b) MoO_2(110), c) RuO_2(110), and d) PtO_2(110) surfaces. The BLYP functional (empty green triangle) and PBEsol functional (empty orange triangle) are shown for a) TiO_2(110) and d) PtO_2(110). Symbols are colored according to tuning parameters: α_{HF} tuning (circles) from 0% (white) to 50% (blue). The DLPNO-CCSD(T) reference is shown with a gray diamond annotated "REF", as described in the main text, for selected a) TiO_2(110) and d) PtO_2(110).

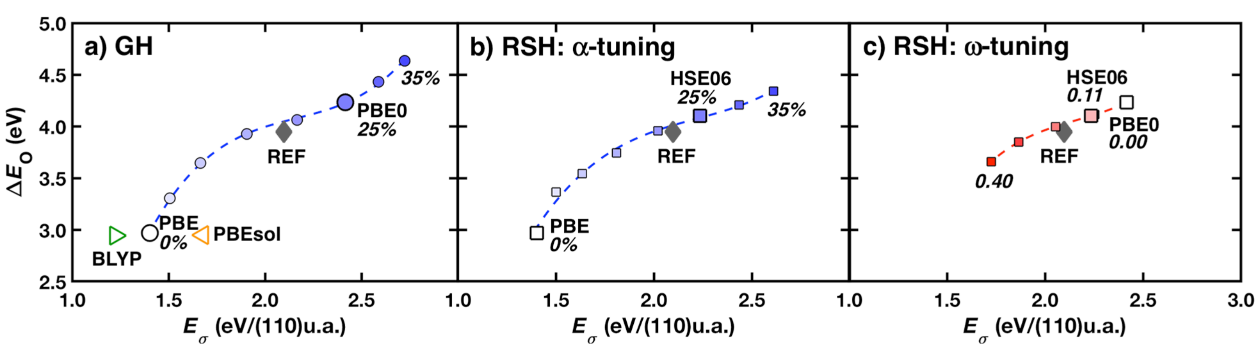

Figure S10. PtO_2(110) E_σ in eV/(110) unit area (u.a.) versus ΔE_O in eV for a) α_{HF} tuning the PBE0 global hybrid (GH) functional, b) α_{HF}-tuning, or c) ω-tuning the HSE06 range-separated hybrid (RSH) functional. Symbols are colored according to tuning parameters: α_{HF} tuning (circles) from 0% (white) to 35% (blue), ω tuning (squares) from 0.00 bohr^{-1} (white) to 0.40 bohr^{-1} (red). The BLYP (empty green triangle) or PBEsol (empty orange triangle) result is added for comparison, and the DLPNO-CCSD(T) reference is shown with a gray diamond annotated "REF".

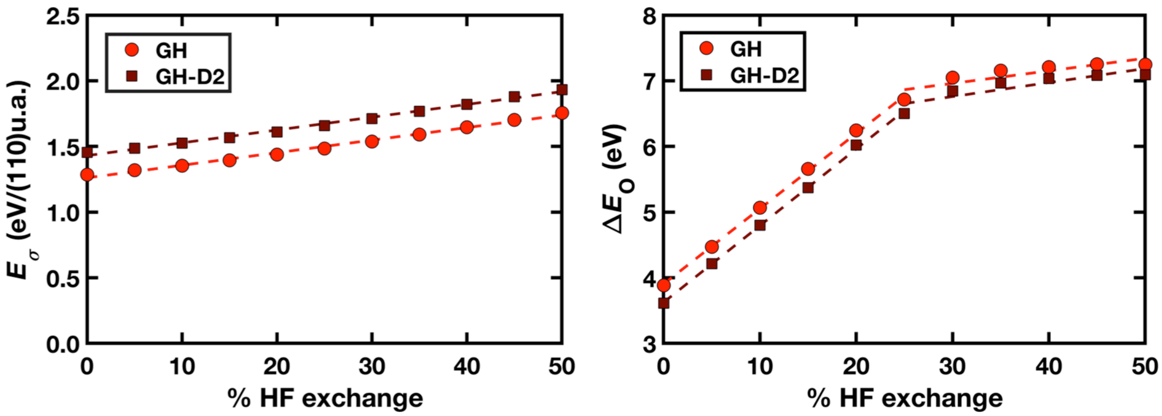

Figure S11. Comparison of surface formation energies, E_σ, in eV/(110) unit area (u.a.) (left), and oxygen adsorption energies, ΔE_O, in eV (right) with percentage of HF exchange for TiO_2 without (red circles) and with D2 dispersion correction[1] (maroon squares). The dashed lines indicate linear best fits through all data to quantify the surface formation energy sensitivities or oxygen adsorption energy sensitivities to HF exchange.

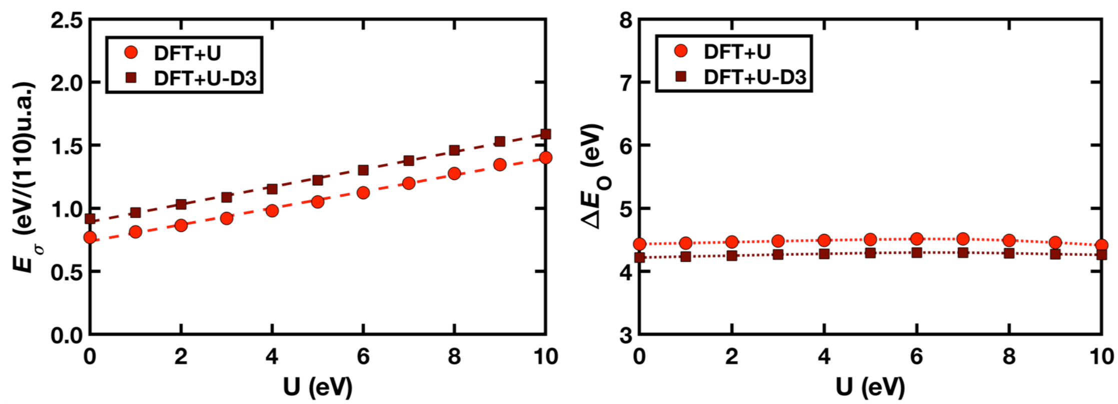

Figure S12. Comparison of surface formation energies, E_σ, in eV/(110) unit area (u.a.) (left), and oxygen adsorption energies, ΔE_O, in eV (right) with U (in eV) for TiO_2 without (red circles) and with D3 dispersion correction[2] (maroon squares). The dashed lines indicate linear best fits through all data to quantify the surface formation energy sensitivities or oxygen adsorption energy sensitivities to U.

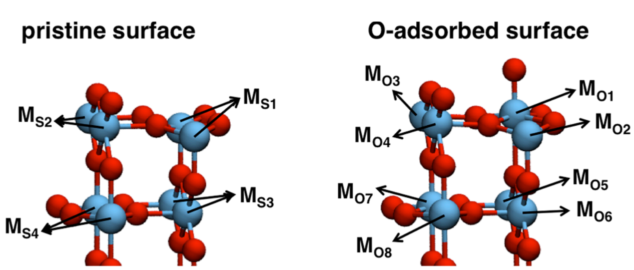

Figure S13. Position of unique metal atoms in the pristine surface and O-adsorbed rutile surface.

Table S7. Occupation matrix, **n,** for use in the "+U" correction for Ti 3*d* orbitals in the TiO_2 bulk structure. In the bulk structure, there are two unique metal atoms, M_{b1} and M_{b2}, in each unit cell.

M_{b1}					M_{b2}				
0.261	0.000	0.000	0.000	-0.092	0.261	0.000	0.000	0.000	0.092
0.000	0.326	-0.098	0.000	0.000	0.000	0.326	0.098	0.000	0.000
0.000	-0.098	0.326	0.000	0.000	0.000	0.098	0.326	0.000	0.000
0.000	0.000	0.000	0.223	0.000	0.000	0.000	0.000	0.223	0.000
-0.092	0.000	0.000	0.000	0.336	0.092	0.000	0.000	0.000	0.336

Table S8. Occupation matrix, **n,** for use in the "+U" correction for Ti 3*d* orbitals in the TiO_2 pristine surface structure. In the pristine surface structure, there are four unique metal atoms, M_{S1}, M_{S2}, M_{S3}, and M_{S4}, in each cell.

M_{S1}					M_{S2}				
0.345	0.000	0.000	-0.025	0.000	0.263	0.000	0.000	0.030	0.000
0.000	0.252	0.000	0.000	0.000	0.000	0.427	0.000	0.000	0.000
0.000	0.000	0.248	0.000	0.000	0.000	0.000	0.195	0.000	0.000
-0.025	0.000	0.000	0.203	0.000	0.030	0.000	0.000	0.366	0.000
0.000	0.000	0.000	0.000	0.431	0.000	0.000	0.000	0.000	0.236
M_{S3}					M_{S4}				
0.234	0.000	0.000	0.082	0.000	0.412	0.000	0.000	-0.014	0.000
0.000	0.414	0.000	0.000	0.000	0.000	0.231	0.000	0.000	0.000
0.000	0.000	0.234	0.000	0.000	0.000	0.000	0.232	0.000	0.000
0.082	0.000	0.000	0.364	0.000	-0.014	0.000	0.000	0.194	0.000
0.000	0.000	0.000	0.000	0.227	0.000	0.000	0.000	0.000	0.415

Table S9. Occupation matrix, **n,** for use in the "+U" correction for Ti 3*d* orbitals in the TiO_2 surface with an O-atom adsorbate. In the adsorbate structure, there are eight unique metal atoms, M_{O1}, M_{O2}, M_{O3}, M_{O4}, M_{O5}, M_{O6}, M_{O7}, and M_{O8}, in each cell. The M_{O1} site is where the O atom adsorbs, and M_{O2}, M_{O3}, and M_{O4} are in the same surface layer.

M_{O1}					M_{O2}				
0.436	0.000	0.000	-0.018	0.000	0.353	0.000	0.000	-0.013	0.000
0.000	0.308	0.000	0.000	0.000	0.000	0.257	0.000	0.000	0.000
0.000	0.000	0.301	0.000	0.000	0.000	0.000	0.261	0.000	0.000
-0.018	0.000	0.000	0.136	0.000	-0.013	0.000	0.000	0.199	0.000
0.000	0.000	0.000	0.000	0.346	0.000	0.000	0.000	0.000	0.418
M_{O3}					M_{O4}				
0.262	0.001	0.000	0.036	0.000	0.262	-0.001	0.000	0.036	0.000
0.001	0.412	0.000	-0.001	0.000	-0.001	0.412	0.000	0.001	0.000
0.000	0.000	0.202	0.000	0.000	0.000	0.000	0.202	0.000	0.000
0.036	-0.001	0.000	0.382	0.000	0.036	0.001	0.000	0.382	0.000
0.000	0.000	0.000	0.000	0.235	0.000	0.000	0.000	0.000	0.235
M_{O5}					M_{O6}				
0.246	0.005	0.000	0.072	0.000	0.246	-0.005	0.000	0.072	0.000
0.005	0.421	0.000	-0.016	0.000	-0.005	0.421	0.000	0.016	0.000
0.000	0.000	0.225	0.000	-0.013	0.000	0.000	0.225	0.000	0.013
0.072	-0.016	0.000	0.362	0.000	0.072	0.016	0.000	0.362	0.000
0.000	0.000	-0.013	0.000	0.228	0.000	0.000	0.013	0.000	0.228
M_{O7}					M_{O8}				
0.400	0.000	0.000	-0.014	0.000	0.399	0.000	0.000	-0.014	0.000
0.000	0.239	0.000	0.000	0.000	0.000	0.241	0.000	0.000	0.000
0.000	0.000	0.242	0.000	0.000	0.000	0.000	0.243	0.000	0.000
-0.014	0.000	0.000	0.187	0.000	-0.014	0.000	0.000	0.189	0.000
0.000	0.000	0.000	0.000	0.408	0.000	0.000	0.000	0.000	0.409

Table S10. Fractionality, Tr[**n**(1-**n**)] in a single spin channel, i.e., PBE magnitude of the "+U" correction for 1 eV U applied based on the Ti 3*d* orbitals in TiO_2. This is computed for each of the unique sites in the bulk, pristine surface, or the O-adsorbed surface. The effect of the shift in fractionality in the surface formation energy with increasing U is the difference between the pristine surface fractionalities and bulk fractionalities for the unique surface sites multiplied by a factor of 2 overall (x4 for the number of atoms in the cell, x2 for the number of spin channels, divide by 2 for the factor in the U correction (for sensitivity estimations), and divided by 2 since there are two surfaces): 0.07 eV/eV of U estimated from these PBE values. The effect of the shift in fractionality in the adsorption energy is the difference between the adsorbate surface fractionalities and surface fractionalities for the unique adsorbate sites with no multiplication factor (x2 for the number of spin channels, divide by 2 for the U correction). This value (0.037 eV/eV of U) is larger than expected due to non-linear dependence of ΔE_O on U.

bulk			pristine surface				
M_{b1}	M_{b2}		M_{S1}	M_{S2}	M_{S3}	M_{S4}	
0.9926	0.9926		1.0067	1.0060	0.9946	0.9968	
O-adsorbed surface							
M_{O1}	M_{O2}	M_{O3}	M_{O4}	M_{O5}	M_{O6}	M_{O7}	M_{O8}
1.0126	1.0146	1.0101	1.0101	0.9993	0.9993	0.9985	1.0013

Table S11. Occupation matrix, **n,** for use in the "+U" correction for Pt 5*d* orbitals in the PtO_2 bulk structure. In the bulk structure, there are two unique metal atoms, M_{b1} and M_{b2}, in each unit cell.

M_{b1}					M_{b2}				
0.889	0.000	0.000	0.000	0.131	0.889	0.000	0.000	0.000	-0.131
0.000	0.837	0.131	0.000	0.000	0.000	0.837	-0.131	0.000	0.000
0.000	0.131	0.837	0.000	0.000	0.000	-0.131	0.837	0.000	0.000
0.000	0.000	0.000	0.968	0.000	0.000	0.000	0.000	0.968	0.000
0.131	0.000	0.000	0.000	0.796	-0.131	0.000	0.000	0.000	0.796

Table S12. Occupation matrix, **n,** for use in the "+U" correction for Pt 5*d* orbitals in the PtO_2 pristine surface structure. In the pristine surface structure, there are four unique metal atoms, M_{S1}, M_{S2}, M_{S3}, and M_{S4}, in each cell.

M_{S1}					M_{S2}				
0.760	0.000	0.000	0.028	0.000	0.942	0.000	0.000	-0.095	0.000
0.000	0.955	0.000	0.000	0.000	0.000	0.718	0.000	0.000	0.000
0.000	0.000	0.972	0.000	0.000	0.000	0.000	0.953	0.000	0.000
0.028	0.000	0.000	0.977	0.000	-0.095	0.000	0.000	0.760	0.000
0.000	0.000	0.000	0.000	0.674	0.000	0.000	0.000	0.000	0.957
M_{S3}					M_{S4}				
0.927	0.000	0.000	-0.104	0.000	0.702	0.000	0.000	0.025	0.000
0.000	0.716	0.000	0.000	0.000	0.000	0.971	0.000	0.000	0.000
0.000	0.000	0.969	0.000	0.000	0.000	0.000	0.967	0.000	0.000
-0.104	0.000	0.000	0.753	0.000	0.025	0.000	0.000	0.979	0.000
0.000	0.000	0.000	0.000	0.969	0.000	0.000	0.000	0.000	0.707

Table S13. Occupation matrix, **n,** for use in the "+U" correction for Pt 5*d* orbitals in the PtO_2 surface with an O-atom adsorbate. In the adsorbate structure, there are eight unique metal atoms, M_{O1}, M_{O2}, M_{O3}, M_{O4}, M_{O5}, M_{O6}, M_{O7}, and M_{O8}, in each cell. The M_{O1} site is where the O atom adsorbs, and M_{O2}, M_{O3}, and M_{O4} are in the same surface layer.

M_{O1}					M_{O2}				
0.738	0.000	0.000	0.022	0.000	0.752	0.000	0.000	0.027	0.000
0.000	0.931	0.000	0.000	0.000	0.000	0.958	0.000	0.000	0.000
0.000	0.000	0.922	0.000	0.000	0.000	0.000	0.964	0.000	0.000
0.022	0.000	0.000	0.982	0.000	0.027	0.000	0.000	0.975	0.000
0.000	0.000	0.000	0.000	0.731	0.000	0.000	0.000	0.000	0.686
M_{O3}					M_{O4}				
0.944	0.000	0.000	-0.092	0.000	0.944	0.000	0.000	-0.092	0.000
0.000	0.704	0.000	0.000	0.000	0.000	0.704	0.000	0.000	0.000
0.000	0.000	0.950	0.000	-0.001	0.000	0.000	0.950	0.000	0.001
-0.092	0.000	0.000	0.757	0.000	-0.092	0.000	0.000	0.757	0.000
0.000	0.000	-0.001	0.000	0.955	0.000	0.000	0.001	0.000	0.955
M_{O5}					M_{O6}				
0.932	0.005	0.000	-0.104	0.000	0.932	-0.005	0.000	-0.104	0.000
0.005	0.710	0.000	0.008	0.000	-0.005	0.710	0.000	-0.008	0.000
0.000	0.000	0.967	0.000	0.001	0.000	0.000	0.967	0.000	-0.001
-0.104	0.008	0.000	0.754	0.000	-0.104	-0.008	0.000	0.754	0.000
0.000	0.000	0.001	0.000	0.967	0.000	0.000	-0.001	0.000	0.967
M_{O7}					M_{O8}				
0.702	0.000	0.000	0.025	0.000	0.703	0.000	0.000	0.025	0.000
0.000	0.969	0.000	0.000	0.000	0.000	0.970	0.000	0.000	0.000
0.000	0.000	0.966	0.000	0.000	0.000	0.000	0.965	0.000	0.000
0.025	0.000	0.000	0.979	0.000	0.025	0.000	0.000	0.979	0.000
0.000	0.000	0.000	0.000	0.709	0.000	0.000	0.000	0.000	0.708

Table S14. Fractionality, Tr[**n**(1-**n**)], in a single spin channel, i.e., the PBE magnitude of the "+U" correction for 1 eV U applied based on the Pt 3*d* orbitals in PtO_2. This is computed for each of the unique sites in the bulk, pristine surface, or the O-adsorbed surface. The effect of the shift in fractionality in the surface formation energy with increasing U is the difference between the pristine surface fractionalities and bulk fractionalities for the unique surface sites multiplied by a factor of 2 overall (x4 for the number of atoms in the cell, x2 for the number of spin channels, divide by 2 for the factor in the U correction (for sensitivity estimations), and divided by 2 since there are two surfaces): 0.006 eV/eV of U estimated from these PBE values. The effect of the shift in fractionality in the adsorption energy is the difference between the adsorbate surface fractionalities and surface fractionalities for the unique adsorbate sites with no multiplication factor (x2 for the number of spin channels, divide by 2 for the U correction). This value (0.086 eV/eV of U) is larger than expected due to non-linear dependence of ΔE_O on U.

bulk			pristine surface				
M_{b1}	M_{b2}		M_{S1}	M_{S2}	M_{S3}	M_{S4}	
0.4963	0.4963		0.4932	0.5074	0.4955	0.4957	
O-adsorption surface							
M_{O1}	M_{O2}	M_{O3}	M_{O4}	M_{O5}	M_{O6}	M_{O7}	M_{O8}
0.5429	0.4998	0.5187	0.5187	0.4968	0.4968	0.4977	0.4977

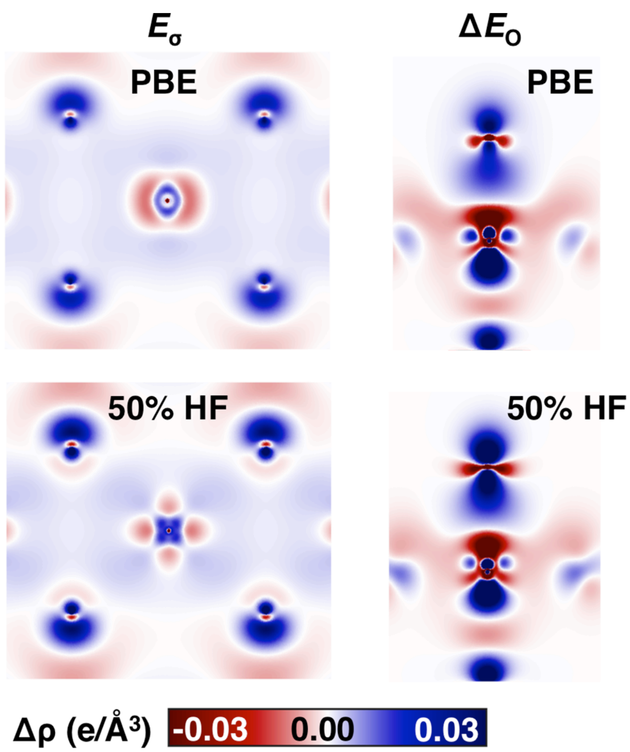

Figure S14. Electron density difference of TiO_2 between the pristine surface and bulk structure at (top left) 0% HF exchange and (bottom left) 50% HF exchange, between O-atom adsorption and the pristine surface with an isolated O atom at (top right) 0% HF exchange and (bottom right) 50% HF exchange. Red and blue colors represent negative (density loss) and positive (density gain) electron density differences, respectively, as indicated by the color bar at bottom.

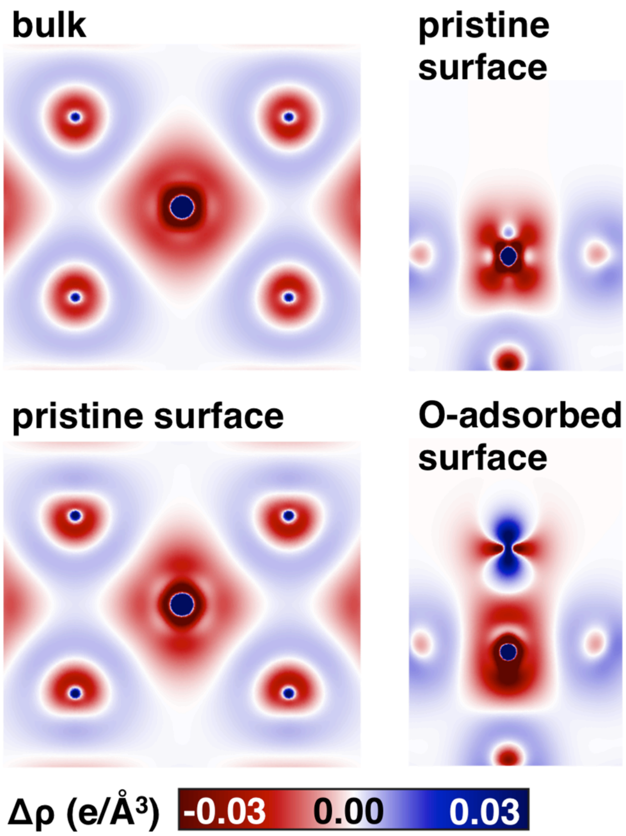

Figure S15. Electron density difference of TiO_2 between 50% and 0% HF exchange for bulk structure (top left), top view pristine surface (bottom left), front view pristine surface (top right), and O-atom adsorption (bottom right). Red and blue colors represent negative (density loss) and positive (density gain) electron density differences, respectively, as indicated by the color bar at bottom.

Table S15. Metal partial charge difference, Δq(M), between pristine surface and bulk structure with increasing fractions of HF exchange. Negative values refer to the metal charge being less positive on the surface than in the bulk. Missing values correspond to calculations at high HF exchange that did not converge.

	PBE	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
TiO_2	-0.07	-0.08	-0.09	-0.09	-0.10	-0.11	-0.12	-0.12	-0.13	-0.13	-0.14
VO_2	-0.05	-0.05	-0.06	-0.06	-0.07	-0.07	-0.08	-0.09	-0.09	-0.09	--
MoO_2	-0.08	-0.08	-0.09	-0.10	-0.10	-0.11	-0.12	-0.12	-0.12	-0.13	-0.13
RuO_2	-0.09	-0.10	-0.10	-0.11	-0.12	-0.12	-0.13	-0.13	-0.14	-0.15	-0.15
RhO_2	-0.04	-0.04	-0.05	-0.06	-0.06	-0.07	-0.07	-0.08	--	--	--
IrO_2	-0.06	-0.07	-0.07	-0.08	-0.08	-0.09	-0.10	-0.10	-0.11	--	--
PtO_2	-0.04	-0.05	-0.05	-0.06	-0.06	-0.06	-0.07	-0.07	-0.08	--	--

Table S16. Metal partial charge difference, Δq(M), between pristine surface and bulk structure with increasing U values (in eV) in the +U correction. Negative values refer to the metal charge being less positive on the surface than in the bulk. Missing values correspond to calculations at high U values that did not converge.

	PBE	1 eV	2 eV	3 eV	4 eV	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
TiO_2	-0.06	-0.06	-0.07	-0.08	-0.09	-0.09	-0.09	-0.10	-0.11	-0.11	-0.11
VO_2	-0.04	-0.04	-0.05	-0.05	-0.06	-0.07	-0.07	-0.08	-0.09	--	--
MoO_2	-0.08	-0.08	-0.08	-0.08	-0.08	-0.09	-0.09	-0.08	-0.08	--	--
RuO_2	-0.10	-0.10	-0.10	-0.11	-0.11	-0.10	-0.10	-0.09	-0.09	-0.08	-0.08
RhO_2	-0.11	-0.11	-0.11	-0.11	-0.11	-0.12	-0.12	-0.12	-0.12	-0.12	-0.11
IrO_2	-0.18	-0.18	-0.18	-0.18	-0.19	-0.19	-0.19	-0.19	-0.19	-0.19	-0.19
PtO_2	-0.22	-0.22	-0.22	-0.22	-0.22	-0.22	-0.21	-0.21	-0.20	-0.20	-0.20

Table S17. Electron density at the adsorbate–surface bond critical point (e/a.u.^{3}) with increasing HF exchange. Missing values correspond to calculations at high HF exchange that did not converge.

	PBE	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
TiO_2	0.151	0.148	0.144	0.141	0.137	0.13	0.115	0.103	0.092	0.081	0.072
VO_2	0.282	0.276	0.271	0.266	0.262	0.257	0.252	0.248	0.244	--	--
MoO_2	0.279	0.272	0.265	0.259	0.253	0.248	0.244	0.240	0.235	--	--
RuO_2	0.242	0.238	0.233	0.229	0.224	0.219	0.215	0.209	0.200	--	--
RhO_2	0.193	0.190	0.187	0.185	0.182	0.176	0.171	--	--	--	--
IrO_2	0.263	0.259	0.255	0.252	0.249	0.246	0.244	0.241	--	--	--
PtO_2	0.178	0.174	0.171	0.168	0.165	0.162	0.158	0.153	--	--	--

Table S18. Electron density at the adsorbate–surface bond critical point (e/a.u.^{3}) with increasing U value (in eV) in the +U correction. Missing values correspond to calculations at high U value that did not converge.

	PBE	1 eV	2 eV	3 eV	4 eV	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
TiO_2	0.121	0.121	0.120	0.121	0.199	0.120	0.121	0.120	0.120	0.121	0.120
VO_2	0.238	0.237	0.237	0.237	0.238	0.236	0.238	0.238	0.238	--	--
MoO_2	0.229	0.226	0.223	0.221	0.218	0.215	0.213	0.210	0.207	--	--
RuO_2	0.187	0.184	0.180	0.176	0.173	0.170	0.168	0.166	0.164	0.163	0.161
RhO_2	0.145	0.142	0.138	0.134	0.130	0.127	0.124	0.121	0.192	0.190	0.189
IrO_2	0.201	0.199	0.197	0.194	0.191	0.188	0.185	0.182	0.179	0.176	0.172
PtO_2	0.136	0.134	0.133	0.131	0.129	0.127	0.125	0.124	0.122	0.120	0.118

Table S19. Oxygen adsorbate partial charge with increasing HF exchange. Missing values correspond to calculations at high HF exchange that did not converge.

	PBE	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%
TiO_2	0.59	0.58	0.57	0.56	0.54	0.46	0.37	0.26	0.18	0.12	0.06
VO_2	0.49	0.47	0.45	0.43	0.42	0.41	0.40	0.39	0.39	--	--
MoO_2	0.53	0.51	0.49	0.48	0.46	0.44	0.42	0.41	0.41	--	--
RuO_2	0.44	0.43	0.42	0.41	0.39	0.38	0.36	0.35	0.33	--	--
RhO_2	0.47	0.46	0.45	0.43	0.42	0.40	0.38	--	--	--	--
IrO_2	0.42	0.40	0.38	0.37	0.36	0.36	0.35	0.34	--	--	--
PtO_2	0.37	0.35	0.34	0.33	0.32	0.31	0.30	0.29	--	--	--

Table S20. Oxygen adsorbate partial charge with increasing U values. Missing values correspond to calculations at high U value that did not converge.

	PBE	1 eV	2 eV	3 eV	4 eV	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
TiO_2	0.57	0.58	0.57	0.59	0.58	0.59	0.59	0.59	0.57	0.57	0.56
VO_2	0.52	0.53	0.52	0.52	0.51	0.51	0.51	0.52	0.51	--	--
MoO_2	0.54	0.54	0.55	0.54	0.53	0.54	0.54	0.54	0.54	--	--
RuO_2	0.50	0.48	0.47	0.46	0.44	0.43	0.42	0.41	0.40	0.39	0.39
RhO_2	0.47	0.46	0.45	0.43	0.42	0.41	0.41	0.40	0.40	0.39	0.39
IrO_2	0.45	0.44	0.43	0.42	0.40	0.39	0.38	0.36	0.35	0.34	0.33
PtO_2	0.47	0.45	0.44	0.43	0.41	0.40	0.39	0.39	0.37	0.37	0.36

Table S21. List of basis set used in this work obtained from the CRYSTAL website. At least a double-ζ split-valence basis set was used for all atoms, triple-ζ split-valence basis sets were used for Pt and Ir, and polarization functions were added for Ir. All-electron calculations were carried out for all atoms except Mo, Ir, and Pt for which Hay-Wadt[3] effective core potentials (ECPs) were used with semicore states of Mo (4*s*, 4*p*) and Ir and Pt (5*s*, 5*p*) treated in the valence. For heavy metals Pt and Ir, as well as Mo, the basis sets used in this work are the only available basis sets on the CRYSTAL website. For other rutile-type transition metal dioxides MO_2 with available experimental crystal structures (M = Ti, V, Ru, Rh), the double-ζ split-valence basis set was used for consistency since this is the only available basis set for all materials considered here. For Fe and Ni, double-ζ split-valence plus polarization basis set was available and used.

Element	**Basis set**
Ti	double-ζ split-valence, all electrons (Ti_86-411(d31)G_darco_unpub)
V	double-ζ split-valence, all electrons (V_86-411d31G_harrison_1993)
Mo	double-ζ split-valence, effective core potentials (Mo_SC_HAYWSC-311(d31)G)
Fe	double-ζ split-valence plus polarization, all electrons (POB-DZVP)
Ru	double-ζ split-valence, all electrons (Ru_9763-11d631G_Sharma_2015)
Rh	double-ζ split-valence, all electrons (Rh_All_Electron_kokalj_1998_unpub)
Ir	triple-ζ split-valence plus polarization, effective core potential (Ir_ Pielnhofer_2015)
Ni	double-ζ split-valence plus polarization, all electrons (POB-DZVP)
Pt	triple-ζ split-valence, effective core potentials (Pt_doll_2004)
O	double-ζ split-valence, all electrons (O_8-411d1_bredow_2006)

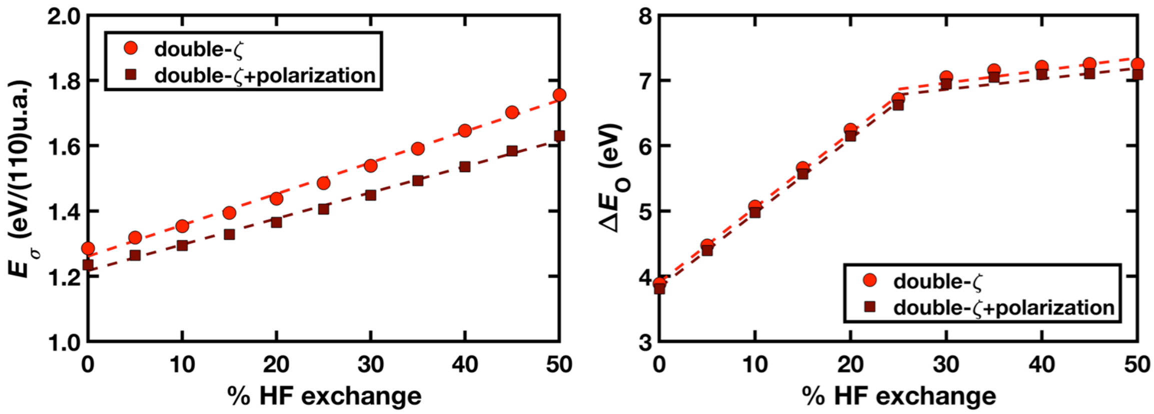

Figure S16. Comparison of surface formation energies, E_σ, in eV/(110) unit area (u.a.) (left), and oxygen adsorption energies, ΔE_O, in eV (right) with percentage of HF exchange for TiO_2 between double-ζ split-valence basis set (red circles) and double-ζ split-valence plus polarization basis set (POB-DZVP, maroon squares) for Ti. The dashed lines indicate linear best fits through all data to quantify the surface formation energy sensitivities or oxygen adsorption energy sensitivities to HF exchange.

Table S22. List of pseudopotentials used in this work with filenames as obtained from the Quantum-ESPRESSO website. (.UPF file extension is omitted).

Element	Pseudopotential name
Ti	Ti. pbe-sp-van_ak
V	V.pbe-sp-van
Mo	Mo.pbe-spn-rrkjus_psl.0.2
Fe	Fe.pbe-sp-van
Ru	Ru.pbe-n-van
Rh	Rh.pbe-rrkjus
Ir	Ir.pbe-n-rrkjus
Ni	Ni.pbe-nd-rrkjus
Pt	Pt.pbe-n-van
O	O.pbe-van_ak

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